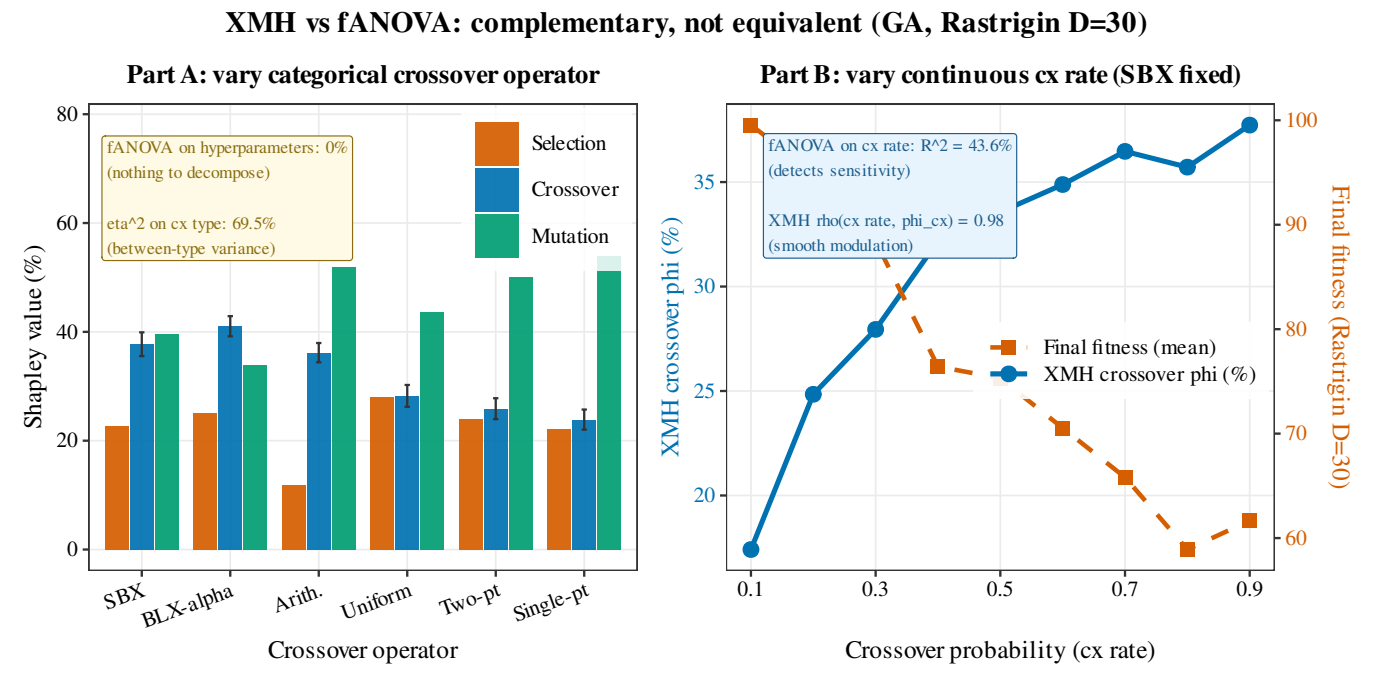
XMH vs fANOVA: complementary, not equivalent (GA, Rastrigin D=30)
Part A: vary categorical crossover operator
fANOVA on hyperparameters: 0%
(nothing to decompose)
eta^2 on cx type: 69.5%
(between-type variance)
Selection
Crossover
Mutation
Shapley value (%)
0
20
40
60
80
SBX
BLX-alpha
Arith.
Uniform
Two-pt
Single-pt
Crossover operator
Part B: vary continuous cx rate (SBX fixed)
fANOVA on cx rate: R^2 = 43.6%
(detects sensitivity)
XMH rho(cx rate, phi_cx) = 0.98
(smooth modulation)
Final fitness (mean)
XMH crossover phi (%)
XMH crossover phi (%)
20
25
30
35
Final fitness (Rastrigin D=30)
60
70
80
90
100
0.1
0.3
0.5
0.7
0.9
Crossover probability (cx rate)